

# Answers to the referee report

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The authors would like to thank the referee for the careful reading of the paper and the valuable comments. We have revised the paper according to the referee's report.

## Answers to the six comments:

1. page 3, paragraph following Definition 1.1:  $\mathcal{L}(u(t, \cdot; \mu)$  is missing a parenthesis on the right

A: We have added the missing right parenthesis for this appearance and a few others.

2. page 5, the reviewer finds it instructive to view “Dalang’s function”  $\Upsilon(\beta)$ , for  $\beta > 0$ , as being an interpolation between its strengthened counterparts, namely:

$$\Upsilon(\beta) \leq 2\Upsilon_\alpha(\beta)^{1/2}\Upsilon_\alpha(0)^{1/2} \quad (0.1)$$

which can be proven by simply splitting the integral into near and far fields, then optimizing the constant. This would complement the discussion on page 5/6, as well as the one in Section 3.2

- A: We thank the referee for bringing up this inequality with an effort to clarify the relation of the standard Dalang’s condition and its strengthened versions. We agree with the referee and indeed, relation (0.1) can be obtained by combining the following two inequalities:

$$\Upsilon(\beta) < \Upsilon(0) \quad \text{and} \quad \Upsilon(\beta) < \beta^\alpha \Upsilon_\alpha(\beta).$$

One can also obtain similar inequality such as

$$\Upsilon(\beta) < \min(\Upsilon(0), \beta^\alpha \Upsilon_\alpha(\beta)).$$

In order to make the relation of the following four conditions clearer:

$$\Upsilon(\beta) < \infty, \quad \Upsilon(0) < \infty, \quad \Upsilon_\alpha(\beta) < \infty, \quad \Upsilon_\alpha(0) < \infty,$$

we have made the following changes:

- (a) We have also included one footnote on p. 6 to provide one example constructed in [CHKK19] that

$$\Upsilon(\beta) < \infty, \quad \Upsilon(0) < \infty, \quad \Upsilon_\alpha(\beta) = \infty, \quad \Upsilon_\alpha(0) = \infty.$$

- (b) We have slightly updated Remark 1.1.  
(c) We have worked out Figure 1.1 with more information.  
(d) We have expanded the discussion in Example 5.10 and included Figure 5.5 to show the subtlety of these conditions.
3. page 7, proof of Corollary 1.2, second equation line: there is a  $*$  missing on  $\zeta$  inside the last integral preceding the inequality. The integrand should be  $(G(t, \cdot) * \rho)(y) |\zeta^*(y)|^2$

A: It has been modified as suggested.

4. page 9, definition of  $H$ : the authors should include the definition of  $H$  since it is invoked several times in the paper and its properties rely on the restrictions associated to the Dalang-type condition. The reviewer went to reference [3] and does not think it will be difficult to add and will only add clarity to their framework.

A: We have introduced Remark 2.1 to define  $H$  and to state its basic properties.

5. page 14, the convention in writing integrals is inconsistent. Although this point may seem pedantic, the authors write integrals in at least three different ways, for example as  $\iint dx dy f(x, y)$  or  $\iint f(x, y) dx dy$  or  $\int dx (f(x, y) dy)$ . The author identified that the main convention seems to be to write integrals as  $\iint dx dy f(x, y)$  when calculations or estimates with integrals are being made, while  $\iint f(x, y) dx dy$  when statements are asserted about them such as conditions being stated. The reviewer suggests that the authors make their convention uniform throughout the paper

A: We have unified the notation for integral throughout the paper to the convention to be  $\iint dx dy f(x, y)$ . All integrals have been updated.

6. page 26, sentence preceding Section 5.4: “Combing” should be “Combining”

A: It has been modified as suggested.

### **Additional changes that we have made**

1. We have expressed our gratitude to the anonymous referee in the Acknowledgments.
2. We have updated the funding information for the first author.
3. The first case in Eq. (5.21) has been updated. This is a copy-paste error during our edit.
4. We have included the table of contents.

## References

- [CHKK19] Le Chen, Jingyu Huang, Davar Khoshnevisan, and Kunwoo Kim. Dense blowup for parabolic SPDEs. *Electron. J. Probab.*, 24:Paper No. 118, 33, 2019.